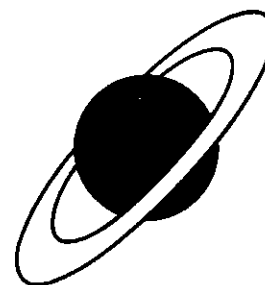


LABORATOIRE NATIONAL SATURNE



91191 Gif-sur-Yvette Cédex France



Spin physics at Saturne II

R.A. Kunne

Laboratoire National SATURNE, CEN-Saclay, 91191 Gif-sur-Yvette Cedex, France

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Laboratoire National SATURNE, CEN-Saclay, 91191 Gif-sur-Yvette Cedex, France

Abstract.

A non-exhaustive overview is given of the spin physics program of the SATURNE II synchrotron.

1. SATURNE II: a unique spin physics tool

The Laboratoire National Saturne, located at 20 km South-West of Paris, is the housing site of the SATURNE II strong focussing synchrotron. The accelerator's diversity, flexibility and performance make it a unique tool for nuclear and intermediary energy particle physics. It is capable of accelerating protons (up to 2.9 GeV) and various ions, ranging from deuterons to iodine ($E_{max} = 1.15$ GeV/nucleon).

As for its polarized beams, SATURNE II is very competitive in its energy domain (table 1). The polarized proton beam holds worldrecords for intensity (3×10^{11} particles per burst) and polarization (decreasing from 90% to 75% with increasing energy).

Table 1

Beams	E_{max} (GeV/nucleon)	Intensity particle/burst	Polarization in %
$\vec{p}$	2.9	3×10^{11}	90 $\rightarrow$ 75
$\vec{d}$	1.15	3×10^{11}	90 $\rightarrow$ 75 (<i>tensor</i>) 60 (<i>vector</i>)
$\vec{n}$	1.15	$3 \times 10^6/cm^2$	60
${}^6\vec{Li}$	1.15	7×10^8	70

Polarized beams available at SATURNE II with their maximum kinetic energy, intensity and polarization. For p and d beams, polarizations at minimum and maximum energy are given. Non-polarized beams that are available as well: 3He , 4He , ${}^{12}C$, ${}^{14}N$, ${}^{16}O$, ${}^{20}Ne$, ${}^{40}Ar$, ${}^{84}Kr$ and ${}^{127}I$.

The deuteron beam has the same maximum intensity as the protons. The vector polarization amounts to 60% and the tensor polarization varies from 90% to 75%. A polarized neutron beam is produced by break-up of the deuterons on a *Be* target, which produces an intense ($3 \times 10^6/cm^2$), monochrome ($dp/p = 3 \times 10^{-2}$) secondary beam. Recently a polarized 6Li beam was used for the first time. Beams are ejected from two extraction points in the ring and may be used at two different energies by two different users.

Six beamlines are permanently occupied, by four spectrometers SPES I to SPES IV, the Nucleon–Nucleon set-up and the large solid angle detector DIOGENE. The neutron beam line feeds the 4π solid angle ARCOLE detector. Five more beam lines may be used for shorter experiments.

Laboratoire National Saturne is an open laboratory, with a user community of about 400 users, half of which are non-French physicists. Experimental proposals should be sent to [1].

2. The physics program

The SATURNE II experiments may be classified in four groups:

- study of the nucleon–nucleon interaction.
- meson production. SATURNE II's four spectrometers are used extensively to study meson production at threshold with various beam and target particles. Special mention should be given to the anomalously large η production in the reaction $pd \rightarrow {}^3He \eta$, which is used as a source of "tagged" η s. An example of a one-pion production study with the ARCOLE detector will be given.
- "pure" nuclear physics. Example: isoscalar-spin transitions.
- strangeness production.

A large number of these experiments use polarized beams as we will see in more detail in the following sections.

2.1. The Nucleon–Nucleon program

The nucleon–nucleon interaction has been studied extensively at SATURNE II over the last ten years by a SACLAY/GENEVA/LNS collaboration. The purpose of the program is the unambiguous reconstruction of all five complex scattering amplitudes, that describe the interaction. To do this a so-called complete set of spin observables has to be measured. Such a complete set requires at least 11 spin observables at a given energy and angle, including triple scattering observables.

The nucleon–nucleon set-up consists of two telescopes of MWPCs. The arms are rotatable and two settings permit to cover the whole angular domain at a given energy. One arm is equipped with a neutron counter consisting of 15 horizontal bars. The other arm contains a large acceptance carbon polarimeter. A low resolution spectrometer and time-of-flight add in the rejection of inelastic and quasi-elastic events. The beam line is equipped with a solenoid to rotate the spin of the beam particle in three directions. The hydrocarbon polarized target has holding magnets permitting two spin directions.

The program consists of three parts, of which two are practically completed. Proton-proton elastic scattering has been studied from 800 MeV up to 2.9 GeV. In the 800 MeV region results are in good agreement with data available from meson factories. A smooth dependance of scattering amplitudes has been observed, except in the region of 2.1 GeV incident proton energy [2]. A further investigation of this region is in progress.

The second part, neutron-proton scattering between 800 MeV and 1100 MeV, benefitted from the intense, polarized neutron beam available since 1988. Data were taken at seven energies between 800 and 1100 MeV. Data analysis is in progress. For more details see [3].

Over the next three years the n–p program will be extended to 2.9 GeV. The neutron beam being limited to a maximum energy of 1.15 GeV, a polarized proton beam and a new polarized ${}^6\text{LiD}$ target will be used. The latter supplies the polarized neutrons, but with a higher polarization and a smaller dilution factor than the deuterated hydrocarbon targets, that have been used in the past. The collaboration will be enlarged by groups from ARGONNE and DUBNA.

2.2. The ARCOLE experiment

The wealth of data gathered in the elastic N–N channels is poorly complemented by the data on the inelastic channels. Above the one pion production threshold, experimental measurement of spin observables is needed to disentangle possible production mechanisms, which seem to be dominated by $N\Delta$ and NN^* intermediate states.

Most existing data on inelastic np scattering were obtained with a deuterium target or beam as neutron source. A SACLAY/LNS collaboration [4] has taken data on the reaction $\bar{n}p \rightarrow pp\pi^-$ (and simultaneously on $\bar{n}p \rightarrow d\pi^+\pi^-$) using SATURNE II's polarized neutron beam. In the set-up shown in figure 1, the neutron beam comes from the left. A LH_2 target is placed in the center of three cylindrical MWPCs with additional cathode strip read-out. The forward region is covered by two plane MWPCs.

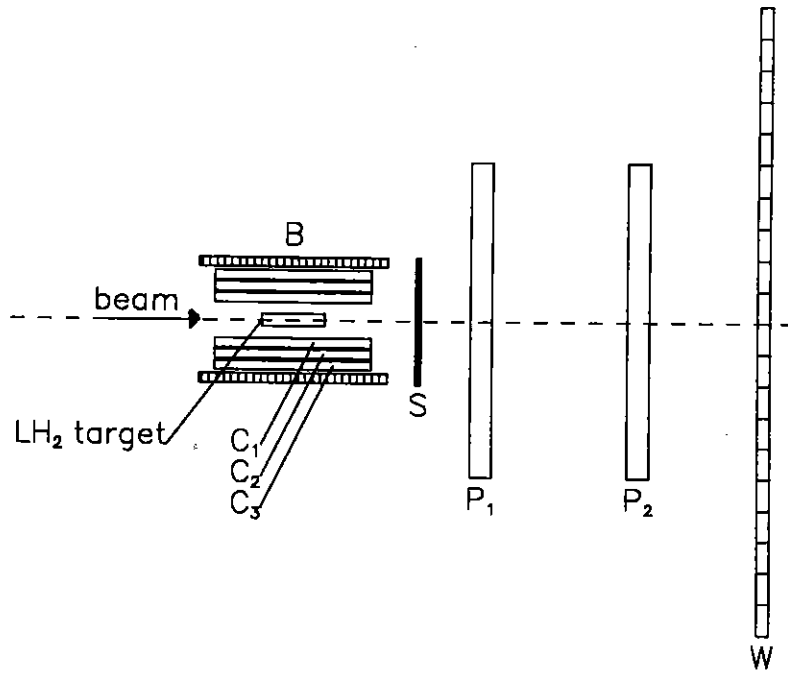


Figure 1. The ARCOLE experimental set-up. *B*: scintillating barrel; *W*: scintillating wall; *S*: start of TOF; C_1 , C_2 , C_3 : cylindrical wire chambers; P_1 , P_2 , P_3 : plane wire chambers.

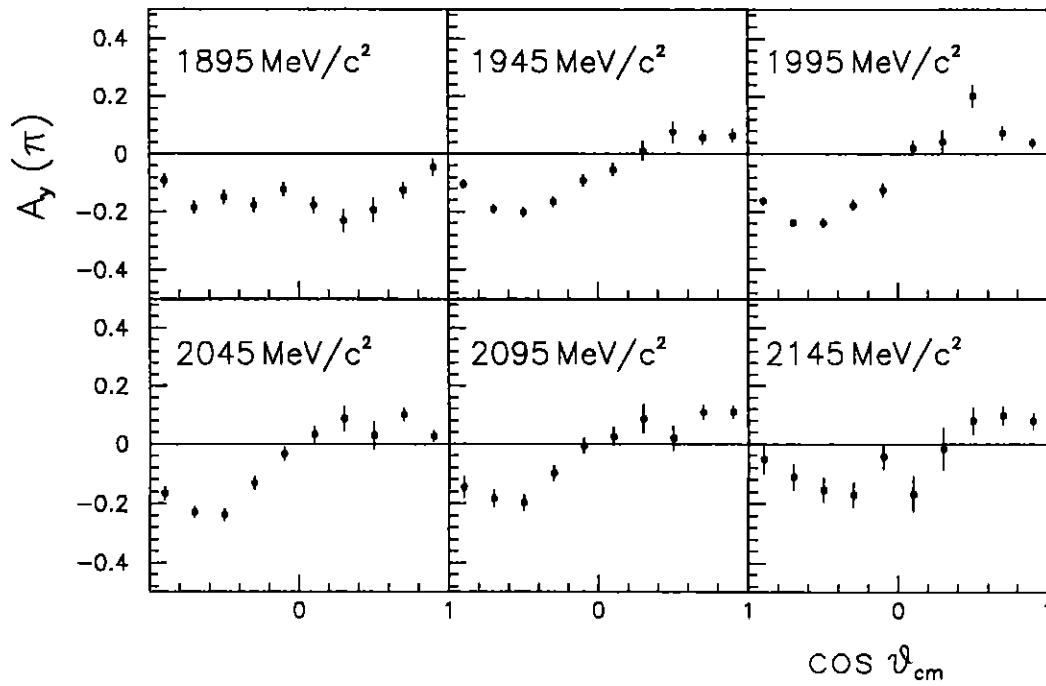


Figure 2. Preliminary results of $A_y(\pi)$ as a function of $\cos \theta_{cm}$ for the reaction $\bar{n}p \rightarrow pp\pi^-$ at $T_{kin} = 1012$ MeV. Six slices of the invariant pp mass are shown.

The trigger system consist of a barrel of eight 1 cm thick plastic scintillators and a $2 \times 2 \text{ m}^2$ plane of twenty 2 cm thick plastic scintillators. Apart from the multiplicity of the event, the scintillators provide dE/dx and TOF information, used to separate protons from pions. The particle momenta are reconstructed from the measured directions. The particle identifications and the total energy constraint are used to eliminate the background. Phase space coverage is almost complete. The absolute neutron flux is determined by means of a separately calibrated neutron monitor behind the set-up, permitting the measurement of differential cross sections, as well as asymmetries.

Data have been taken at 572, 784, 1012 and 1134 incident neutron kinetic energy. In figure 2 preliminary results of the pion asymmetry at 1012 MeV are shown. Close to threshold $A_y(\pi)$ seems to have a different behaviour than at higher values of M_{pp} , a possible indication of final state interaction between the protons.

2.3. Nuclear physics experiments

Nuclear excitation experiments form a large fraction of the SATURNE II physics program. The projectile is used to excite the target nucleus by way of inelastic scattering, charge-exchange or a pick-up reaction. A polarized probe opens the possibility of selecting specific spin states. An example is an experiment by the ORSAY/LNS/RUTGERS/GEORGIA collaboration [5], where $(\vec{d}, \vec{d}')$ scattering is used to probe isoscalar-spin transitions in the nucleus. The polarized deuteron (a $\Delta T = 0$ particle) with an energy of 200 MeV/nucleon excites the $\Delta S = 0, 1$ levels of the target nucleus. The tensor analysing power A_{yy} and the polarization transfer $K_y^{y'}$ are used to form a signature for $\Delta S = 1$ transitions in the following way:

$$S_d^y = \frac{4}{3} + \frac{2}{3} A_{yy} - 2 K_y^{y'}$$

The signature is equivalent to the spinflip probability S_{nn} in $(\vec{p}, \vec{p}')$ scattering. S_d^y is positive for a $\Delta S = 1$ transition and ≈ 0 for a $\Delta S = 0$ transition.

The set-up is shown in figure 3. The deuteron, coming from the right, impinges on the target. The deflected particles are momentum analyzed in the spectrometer and their polarization is measured in a polarimeter. The latter consists of three MWPCs before and after the carbon slab. Deuteron break-up products are stopped in the iron. The trigger is given by the coincidence of scintillators P1, P2 and Q. Events without a scattering in the carbon are rejected on-line.

So far ^{12}C and ^{40}Ca have been investigated. The first target was used as a test case to define the signature [6]. Three $\Delta S = 0$ levels and one $\Delta S = 1$ level were

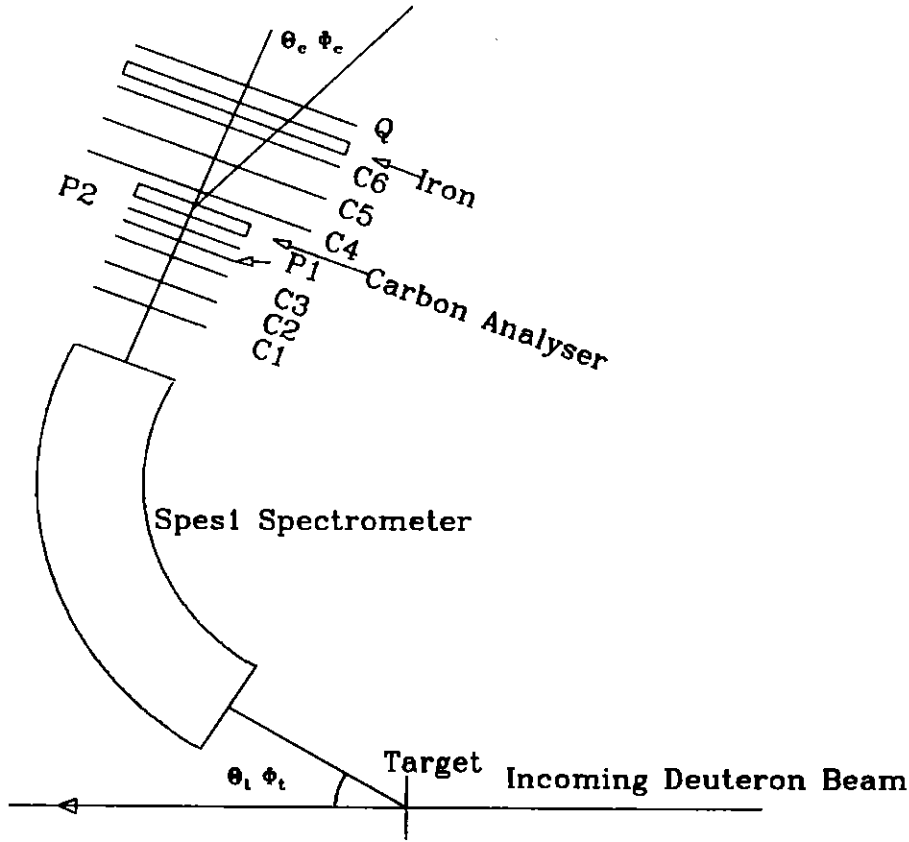


Figure 3. Set-up of the isoscalar-spin transition experiment.

already known, and are clearly identified by the signature (figure 4). Significant $\Delta S = 1$ strength appears also above 18 MeV excitation energy. In ^{40}Ca the continuum has been explored up to 40 MeV excitation energy. In this case the signature has allowed for a separation of the $\Delta S = 0$ and $\Delta S = 1$ parts of the cross section (figure 5). The $\Delta S = 1$ component is 5–10 times smaller than the $\Delta S = 0$ part. The spin force is concentrated around 9 and 15 MeV and seems diluted between 20 and 35 MeV. A comparison with $(\vec{p}, \vec{p}')$ scattering will allow to disentangle the $\Delta S = 0, 1$ and $\Delta T = 0, 1$ components.

2.4. The Strangeness project

It is surprising that, despite the theoretical interest in strangeness production, little is known on the production of strange baryons in nucleon–nucleon interactions. The simplest pp reactions in which a strange baryon is produced, are $pp \rightarrow pK^+\Lambda^0$ and $pp \rightarrow pK^+\Sigma^0$. At the energies that SATURNE II can provide only total cross sections were measured in bubble chamber experiments [7], which collected only a few hundred events below 3 GeV.

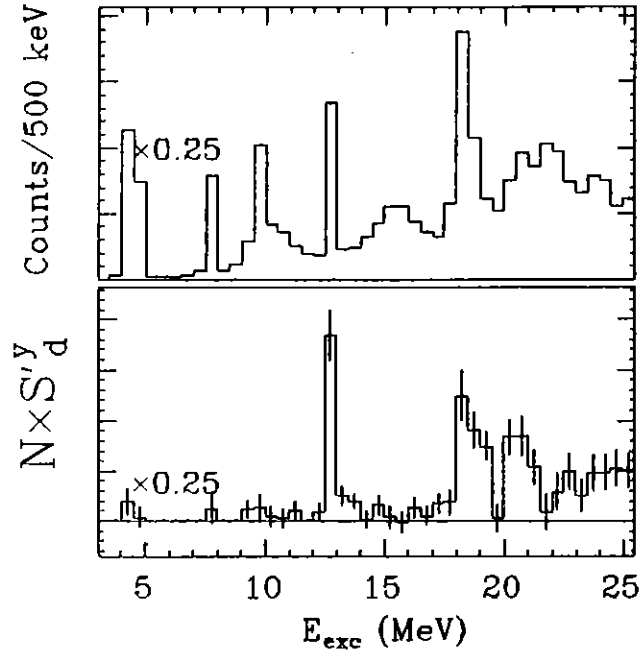


Figure 4. Inelastic ($\vec{d}, \vec{d}'$) spectrum (arbitrary units) of ^{12}C , measured at a laboratory angle of 4° and an incident deuteron energy of 400 MeV. In the lower plot, the spectrum is multiplied by the signature showing the $\Delta S = 1$ transitions.

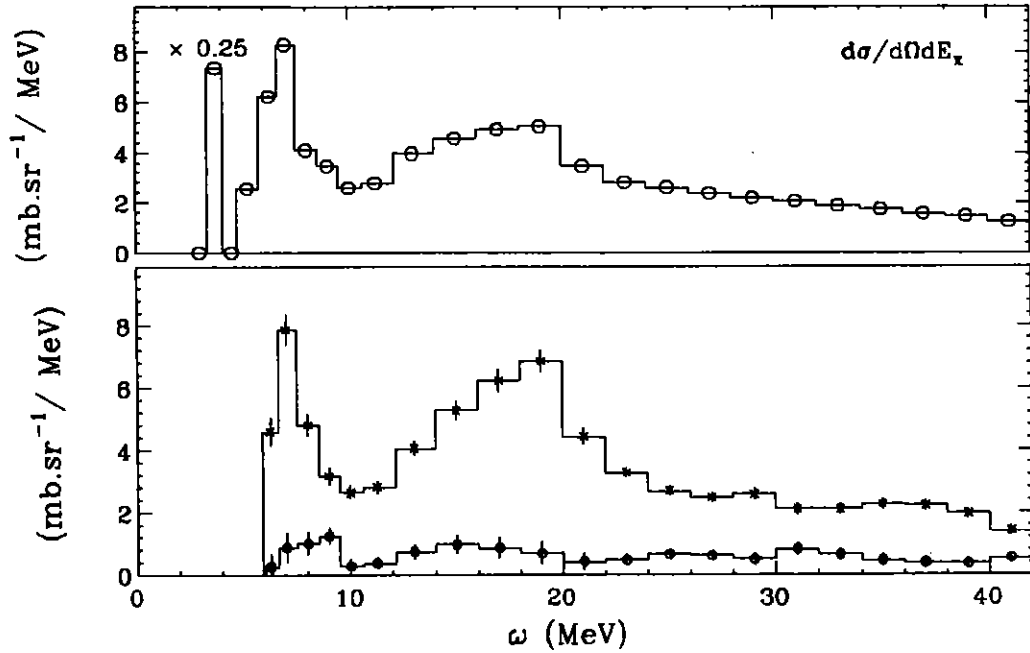


Figure 5. Inelastic ($\vec{d}, \vec{d}'$) spectrum of ^{40}Ca measured at a laboratory angle of 4° and an incident deuteron energy of 400 MeV. In the lower plot, the $\Delta S = 0$ and the $\Delta S = 1$ parts are separated with the aid of the signature.

Recently very accurate, but inclusive kaon production data were obtained at SATURNE II [8]. Differential cross sections and spin observables do not exist at all close to threshold. Despite the lack of knowledge on the elementary hyperon production channels, these data are needed to

- test meson exchange theories in strangeness production. Many models exist to describe YN , KN , KY interactions and the reactions $\gamma N \rightarrow YK$, $\bar{p}p \rightarrow \bar{Y}Y$, $NN \rightarrow NYK$.
- investigate in more detail the hyperon-nucleon interaction, including YN final state interactions and hypernuclear bound states;
- study mechanisms based on the quark model, which is a possibility as the s -quark mass is large compared to m_u , m_d ;
- shed light on the badly understood high polarization of inclusively measured Λ^0 s at energies larger than 10 GeV;
- test nuclear physics theories, that use K^+ as a probe of the conditions inside nuclei, and for which the elementary production reactions form the input.

All these items require the measurement of differential cross section and spin observables in the separated $pp \rightarrow pK^+\Lambda^0$ and $pp \rightarrow pK^+\Sigma^0$ reactions.

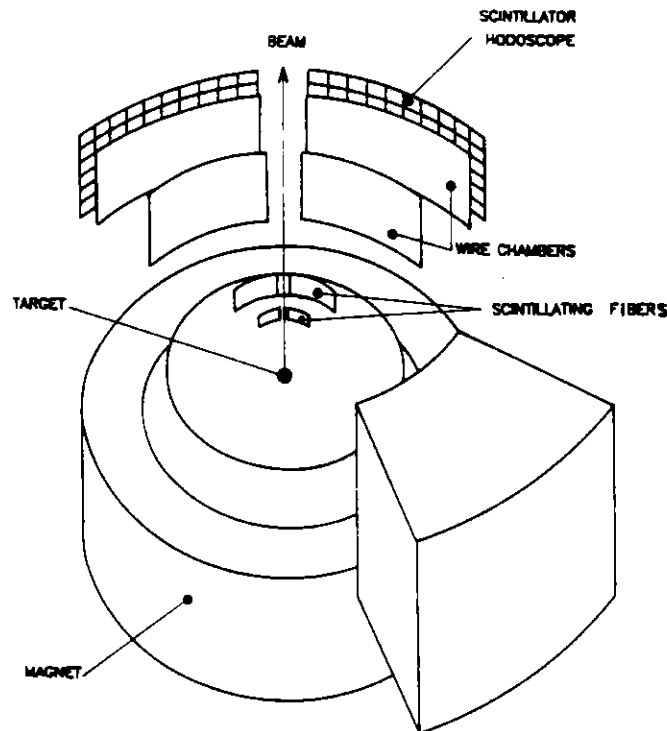


Figure 6. Set-up of the strangeness production experiment.

A TORINO/DUBNA/INDIANA/LNS collaboration [9] plans to fill part of the gap, by measuring (in a first phase) the spin observables P_Λ , P_Σ , A_n and D_{nn} . The proposed set-up is shown in figure 6. The polarized proton beam impinges on a LH_2 target placed in the center of a 1.47 T dipole magnet. A two-armed spectrometer detects the particles. Each arm consists of two scintillator fibres detectors, followed by two MWPCs and a scintillator hodoscope. Each arm has a length of 120 cm and covers a scattering angle of 45° . The dip angle acceptance is $\pm 20^\circ$.

A two-level trigger is proposed to suppress the background of other channels, which together have a cross section that is three orders of magnitude larger than those of the hyperon production channels. The first level is a multiplicity trigger that uses the information from the fiber detectors and scintillator hodoscopes. A second level trigger, using four RISC processors, reconstructs the primary and the Λ^0 decay vertex on-line to reject four prong events from other channels.

The Monte-Carlo simulations show that a clean separation of the two hyperon channels can be achieved with a resolution of about $30 \text{ MeV}/c^2$ FWHM on the mass of the hyperon. The resolution on the proton decay angle in the Λ^0 rest frame (needed for the hyperon polarization measurement) is of the order of one degree (FWHM). The background suppression is better than 10^{-5} .

Assuming 10^8 protons/burst and a 1 cm LH_2 target of the order of 2×10^5 $pK^+\Lambda^0$ and 7×10^4 $pK^+\Sigma^0$ are expected to be collected in three days of data-taking at 2.9 GeV.

3. Conclusions and acknowledgements

SATURNE II is, by far, the best intermediary energy synchrotron in the world as concerning the availability of its polarized beams. The intensity and polarization of the proton, deuteron, neutron and ^6Li beams are unequalled. Consequently it has a steady, large and international user community consisting of nuclear and intermediary energy physicists. A varied physics program includes studies of the fundamental nucleon-nucleon interaction, meson production, strange hyperon production as well as nucleon and nuclear excitation experiments. A large part of the experiments concentrates on spin effects.

The demand for its beams surely shows that, even in a world of diminishing nuclear physics budgets, SATURNE II still earns a distinguished place.

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Laboratoire National Saturne
91191 Gif-sur-Yvette Cedex France
Tél : (1) 69 08 22 03
Téléfax : (1) 69 08 29 70
Bifnet : SATURNE @ FRCPN 11
Télex : ENERG X 604641 F